

Week 1 Python Foundations Lab Report

During this week's Big Data Analytics lab, I learned the basic Python skills needed to work with and analyze data. I practiced using variables and different data types, including strings, integers, floats, and Boolean values. I also learned how to convert data from one type to another.

I worked with strings using f-strings, upper(), len(), and slicing. I then used lists to store quiz scores and practiced calculating the average, highest score, lowest score, and range. I also learned how to add and modify values in a list.

For dictionaries, I created a dictionary containing information about myself and practiced accessing, adding, and updating values. Finally, I used these skills to analyze a dataset of 20 AUCA students. I calculated the class average, identified the highest and lowest scores, counted students by district, and found the top-scoring student.

One thing that surprised me was that "3" + "4" produces "34" instead of 7. I learned that this happens because both values are strings, so Python joins them as text instead of adding them as numbers. This helped me understand why data types are important when programming.

Overall, the lab gave me a good foundation in Python and showed me how basic Python structures can be used to organize and analyze data.